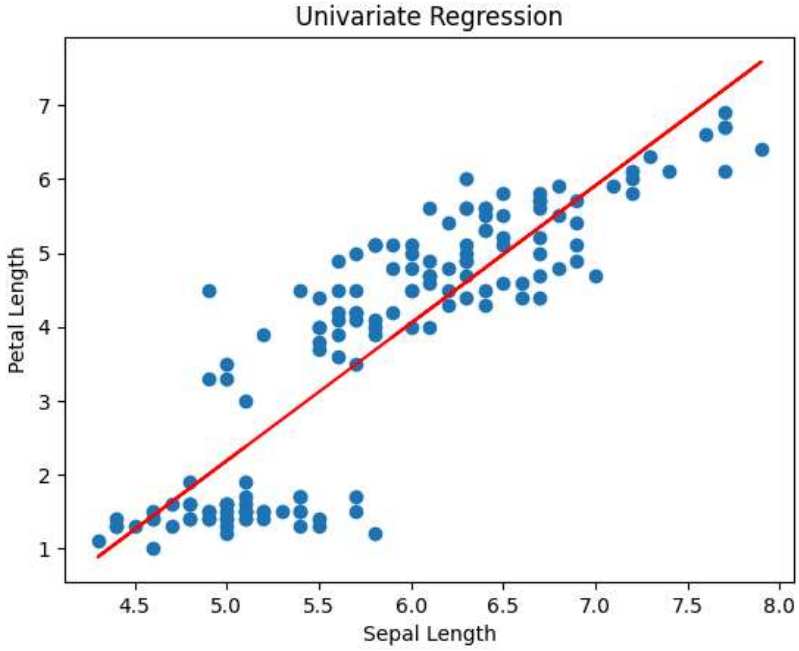
```
# Step 1: Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
from mpl_toolkits.mplot3d import Axes3D

# Step 2: Load dataset
# You can use local file OR sklearn dataset

from sklearn.datasets import load_iris
iris = load_iris()

# Create DataFrame
df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
df['variety'] = iris.target

# Rename columns for easier access
df.columns = ['sepal_length', 'sepal_width', 'petal_length', 'petal_width', 'variety']

print(df.head())
print("Shape:", df.shape)
```

```
   sepal_length  sepal_width  petal_length  petal_width  variety
0           5.1          3.5           1.4          0.2        0
1           4.9          3.0           1.4          0.2        0
2           4.7          3.2           1.3          0.2        0
3           4.6          3.1           1.5          0.2        0
4           5.0          3.6           1.4          0.2        0
Shape: (150, 5)
```

```
# Step 3: Prepare Data
# Convert variety to numeric (already numeric here)

# -------------------------------
# Step 4: UNIVARIATE REGRESSION
# Example: sepal_length → petal_length
# -------------------------------

X_uni = df[['sepal_length']]
y_uni = df['petal_length']

model_uni = LinearRegression()
model_uni.fit(X_uni, y_uni)

y_pred_uni = model_uni.predict(X_uni)

# R2 Score
r2_uni = r2_score(y_uni, y_pred_uni)

print("\n--- Univariate Regression ---")
print("Coefficient:", model_uni.coef_[0])
print("Intercept:", model_uni.intercept_)
print("R2 Score:", r2_uni)

# Plot
plt.scatter(X_uni, y_uni)
plt.plot(X_uni, y_pred_uni, color='red')
plt.xlabel("Sepal Length")
plt.ylabel("Petal Length")
plt.title("Univariate Regression")
plt.show()
```

```
plt.show()
```

```
--- Univariate Regression ---
Coefficient: 1.8584329782548412
Intercept: -7.101443369602455
R2 Score: 0.759954645772515
```

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# -------------------------------
# Step 5: BIVARIATE REGRESSION
# Example: sepal_length + sepal_width → petal_length
# -------------------------------

X_bi = df[['sepal_length', 'sepal_width']]
y_bi = df['petal_length']

model_bi = LinearRegression()
model_bi.fit(X_bi, y_bi)

y_pred_bi = model_bi.predict(X_bi)

r2_bi = r2_score(y_bi, y_pred_bi)

print("\n--- Bivariate Regression ---")
print("Coefficients:", model_bi.coef_)
print("Intercept:", model_bi.intercept_)
print("R2 Score:", r2_bi)

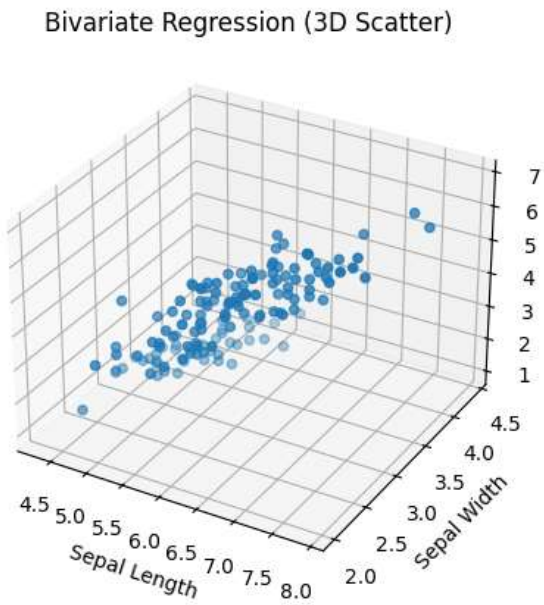
# 3D Plot
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

ax.scatter(df['sepal_length'], df['sepal_width'], y_bi)

ax.set_xlabel('Sepal Length')
ax.set_ylabel('Sepal Width')
ax.set_zlabel('Petal Length')

plt.title("Bivariate Regression (3D Scatter)")
plt.show()
```

```
--- Bivariate Regression ---
Coefficients: [ 1.77559255 -1.33862329]
Intercept: -2.524761511833407
R2 Score: 0.8676860089345194
```

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# -------------------------------
# Step 6: MULTIVARIATE REGRESSION
# Example: all features → petal_length
# -------------------------------

X_multi = df[['sepal_length', 'sepal_width', 'petal_width']]
y_multi = df['petal_length']

model_multi = LinearRegression()
model_multi.fit(X_multi, y_multi)

y_pred_multi = model_multi.predict(X_multi)

r2_multi = r2_score(y_multi, y_pred_multi)

print("\n--- Multivariate Regression ---")
print("Coefficients:", model_multi.coef_)
print("Intercept:", model_multi.intercept_)
print("R2 Score:", r2_multi)

# Plot Actual vs Predicted
plt.scatter(y_multi, y_pred_multi)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Multivariate Regression (Actual vs Predicted)")
plt.show()
```
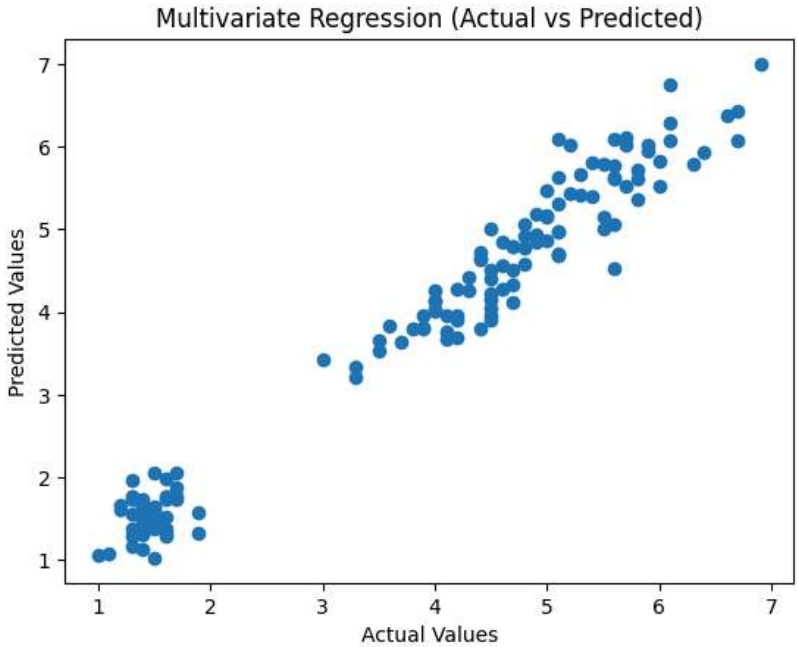
```
--- Multivariate Regression ---
Coefficients: [ 0.72913845 -0.64601244  1.44679341]
Intercept: -0.2627111975741818
R2 Score: 0.9680117693912218
```